

Syed Irfan Ali Meerza

ASSISTANT PROFESSOR @ DEPARTMENT OF COMPUTER SCIENCE, VIRGINIA COMMONWEALTH UNIVERSITY, RICHMOND, VA

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Research Interests

Robust and Trustworthy AI

- Federated Learning
- Machine Unlearning
- Differential Privacy
- Algorithmic Fairness

Responsible Content Governance

- Creative Content Protection
- Generative Model Auditing
- Data Watermarking
- Foundation Model Security

Smart Healthcare and Fitness

- Intelligent Fitness Technologies
- Multimodal Sensing
- Passive Health Tracking
- Early Disease Detection

Education

University of Tennessee, Knoxville

Ph.D. Candidate in Computer Engineering

Advisor: Dr. Jian Liu

Knoxville, TN

Aug 2021–Aug 2026

American International University Bangladesh (AIUB)

M.Sc. in Electrical and Electronics Engineering (**Gold Medalist**)

Dhaka, Bangladesh

Jan 2018–Nov 2019

Khulna University of Engineering and Technology

B.Sc. in Electronics and Communication Engineering

Khulna, Bangladesh

Feb 2011–Jun 2015

Professional Experience

Assistant Professor, Virginia Commonwealth University

Aug. 2026 – Present

Graduate Research Intern, Oak Ridge National Laboratory

Jan. 2025 – May 2025

May 2024 – Aug. 2024

Graduate Research Assistant, University of Tennessee Knoxville

Aug. 2021 – Aug. 2026

Publication

- [1] **[ECCV, 26] Syed Irfan Ali Meerza**, Oktay Ozturk, Amir Sadovnik, Jian Liu, “DiffUE: Enhancing Utility Unlearnability Trade-offs in Unlearnable Examples Against Relearning with Diffusion Autoencoders”. *Proceedings of the 19th European Conference on Computer Vision, (ECCV)*, 2026. (**Acceptance Rate: 27.5%**)
- [2] **[IEEE S&P/Oakland, 26] Syed Irfan Ali Meerza**, Jian Liu, “MusicShield: Protection for Musicians in the Era of Generative AI,” *Proceedings of the 47th IEEE Symposium on Security and Privacy, (IEEE S&P)*, 2026. (**Acceptance Rate: 13%**)

- [3] **[IEEE SaTML, 26] Syed Irfan Ali Meerza**, Feiyi Wang, Jian Liu, “FedSpy-LLM: Towards Scalable and Generalizable Data Reconstruction Attacks from Gradients on LLMs”. *Proceedings of 4th IEEE Conference on Secure and Trustworthy Machine Learning, (SaTML)*, 2026. **(Acceptance Rate: 26%)**
- [4] **[UbiComp, 25] Chandler Jackson Bauder**, Tianhao Wu, **Syed Irfan Ali Meerza**, Aly Fathy, Jian Liu, “mm-RunAssist: mmWave-based Respiratory and Running Rhythm Analysis During Treadmill Workouts,” in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, (IMWUT/UbiComp)*, 2025.
- [5] **[UbiComp, 25] Tianhao Wu**, Yi Wu, Bibek Poudel, **Syed Irfan Ali Meerza**, Rajasi Gore Athawale, Weizi Li, Zan Gao, Cagdas Karatas, Jian Liu, “VibRun: Real-time Unobtrusive Gait Analysis for Treadmill Running via Footstep Vibrations,” in *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies, (IMWUT/UbiComp)*, 2025.
- [6] **[PPAI @ AAAI, 25] Syed Irfan Ali Meerza**, Luyang Liu, Jiaxin Zhang, Jian Liu, “GLocalFair: Jointly Improving Global and Local Group Fairness in Federated Learning,” in *Proceedings of the 6th AAAI Workshop on Privacy-Preserving Artificial Intelligence, (PPAI)*, 2025.
- [7] **[IEEE S&P/Oakland, 25] Syed Irfan Ali Meerza**, Lichao Sun, Jian Liu, “HarmonyCloak: Making Music Audio Unlearnable for Generative AI,” in *Proceedings of the 46th IEEE Symposium on Security and Privacy, (IEEE S&P)*, 2025. **(Acceptance Rate: 14%)**
- [8] **[IJCAI, 24] Syed Irfan Ali Meerza**, Jian Liu, “EAB-FL: Exacerbating Algorithmic Bias Through Model Poisoning Attacks in Federated Learning,” in *Proceedings of the 33rd International Joint Conference on Artificial Intelligence, (IJCAI)*, 2024. **(Acceptance Rate: 14%)**
- [9] **[ISVLSI, 24] Syed Irfan Ali Meerza**, Amir Sadovnik, Jian Liu, “ConFUSE: Confusion-based Federated Unlearning with Saliency Exploration,” in *Proceedings of the IEEE Computer Society Annual Symposium on VLSI, (ISVLSI)*, 2024.
- [10] **[AsiaCCS, 22] Yue Cui**, **Syed Irfan Ali Meerza**, Zhuohang Li, Luyang Liu, Jiaxin Zhang, Jian Liu, “RecUP-FL: Reconciling Utility and Privacy in Federated Learning via User-configurable Privacy Defense,” in *Proceedings of the 18th ACM ASIA Conference on Computer and Communications Security, (AsiaCCS)*, 2022. **(Acceptance Rate: 16%)**
- [11] **[WNYISPW, 22] Afsana Ahamed**, **Syed Irfan Ali Meerza**, “Iris Recognition Using Curvelet Transform and Accuracy Maximization by Particle Swarm Optimization,” in *Proceedings of the IEEE Western New York Image and Signal Processing Workshop, (WNYISPW)*, 2022.
- [12] **[EMBC, 22] Syed Irfan Ali Meerza**, Zhuohang Li, Luyang Liu, Jiaxin Zhang, Jian Liu, “Fair and Privacy-Preserving Alzheimer’s Disease Diagnosis Based on Spontaneous Speech Analysis via Federated Learning,” in *Proceedings of the 44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, (EMBC)*, 2022.
- [13] **[EMBC, 21] Syed Irfan Ali Meerza**, Affan Affan, Hossein Mirinejad, Michael E Brier, Jacek M Zurada, Tamer Inanc, “Precise Warfarin Management Through Personalized Modeling and Control with Limited Clinical Data,” in *Proceedings of the 43rd Annual International Conference of the IEEE Engineering in Medicine and Biology Society, (EMBC)*, 2021.
- [14] **[AAEA, 21] Syed Imran Ali Meerza**, **Syed Irfan Ali Meerza**, Afsana Ahmed, “Food Insecurity Through Machine Learning Lens: Identifying Vulnerable Households,” in *Proceedings of the Agricultural and Applied Economics Association and Western Agricultural Economics Association Joint Annual Meeting, (AAEA/WAEA)*, 2021.
- [15] **[ICASERT, 19] Syed Irfan Ali Meerza**, Moinul Islam, Md Mohiuddin Uzzal, “Q-Learning Based Particle Swarm Optimization Algorithm for Optimal Path Planning of Swarm of Mobile Robots,” in *Pro-*

ceedings of the IEEE 1st International Conference on Advances in Science, Engineering and Robotics Technology, **(ICASERT)**, 2019.

- [16] **[ICREST, 19] Syed Irfan Ali Meerza**, Moinul Islam, Md Mohiuddin Uzzal, “Performance Evaluation of Different Algorithms for Handwritten Isolated Bangla Character Recognition,” in *Proceedings of the 1st International Conference on Robotics, Electrical and Signal Processing Techniques*, **(ICREST)**, 2019.
- [17] **[ICITISEE, 18] Syed Irfan Ali Meerza**, Moinul Islam, Md Mohiuddin Uzzal, “Optimal Path Planning Algorithm for Swarm of Robots Using Particle Swarm Optimization Technique,” in *Proceedings of the IEEE 3rd Conference on Information Technology, Information Systems and Electrical Engineering*, **(ICITISEE)**, 2018.

Manuscripts

- [1] **Syed Irfan Ali Meerza**, Jiawei Yu, Yi Chen Liu, Xi Gua, Jian Liu, “Harmonizing Perception: Human vs. AI Capabilities in Identifying AI-Generated Music”. **(Under review)**
- [2] Jiawei Yu, **Syed Irfan Ali Meerza**, Yi Wu, Amir Sadovnik, Jian Liu, “SemPurify: Semantics-Aware Data Purification Against Backdoor Attacks”. **(Under review)**
- [3] **Syed Irfan Ali Meerza**, Jian Liu, “MusicTrace: Certifiably Robust, General Purpose Watermarking for Traceability Through Generative Music Models”. **(In preparation)**

Patents

- [1] Jian Liu, Syed Irfan Ali Meerza, Lichao Sun, “HarmonyCloak: Making Music Audio Unlearnable for Generative AI,” U.S. Provisional Application, April 2025.
- [2] Jian Liu, Syed Irfan Ali Meerza, “MusicShield: Protection for Musicians in the Era of Generative AI,” U.S. Provisional Application, April 2025.

Awards, Grants and Fellowships

Awards

- “Gonzalez Family Outstanding Graduate Research Assistant,” University of Tennessee, Knoxville, 2025
- “Honorable Mention (Team)”, MagNet Challenge, Princeton University, New Jersey, USA, 2023
- Received “Chancellor’s Award” and “Vice-Chancellor’s Award”, AIUB, 2020
- Winner (Team) (Khulna Division), Digital Innovation Fair, Bangladesh, 2015
- Top 20 nomination for “Young Bangla Youth Award”, Bangladesh, 2015

Grants

- NSF Non-Academic Research Internship for Graduate Students (INTERN) Supplemental Funding (\$55,000), 2024
- GSS Travel Grant UTK (\$1550), 2025
- PPAI Workshop Travel Grant, AAAI Workshop on Privacy-Preserving Artificial Intelligence (\$500), 2025
- GSS Travel Grant UTK (\$1,150), 2024
- IJCAI Travel Grant, 33rd International Joint Conference on Artificial Intelligence (IJCAI) (\$350), 2024

Fellowships

- EERE Fellow, Department of Energy Efficiency and Renewable Energy and University of Tennessee, Knoxville (\$10,000), 2022-2023
- Tennessee's Top 100 Fellow, University of Tennessee, Knoxville (\$40,000), 2021-2025
- Khulna University of Engineering and Technology Merit Scholarship, 2012-2015

Teaching Experience

Lab Instructor, The University of Tennessee, Knoxville Fall 2025

COSC-102 Introduction to Computer Science

Led weekly labs for **35** students; developed exercises and graded assignments.

Teaching Assistant, The University of Tennessee, Knoxville Fall 2024

ECE-569 Mobile and Embedded System Security

Assisted course delivery for **50** graduate students; held office hours and graded projects.

Guest Lecture, The University of Tennessee, Knoxville Spring 2024

COSC-526 Data Mining and Analytics

Delivered a **60-minute** lecture on data privacy and security; received positive feedback from students and instructor.

Teaching Assistant, The University of Tennessee, Knoxville Spring 2024

COSC-526 Data Mining and Analytics

Supported course for **27** graduate students; assisted with grading, projects, and exam preparation.

Teaching Assistant, The University of Tennessee, Knoxville Fall 2023

COSC-522 Machine Learning

Assisted with lectures and assignments for **45** graduate students; held weekly office hours.

Mentorship and Academic Supervision

Undergraduate Students

- Minjae Bae UTK EECS, 2024–Current
- Shawn-Patr Barhorst UTK EECS, 2024–2025
- Maximus Nwider UTK EECS, 2023–2025
- Luis Gonzalez UTK EECS, 2021–2023

MS Students

- Xiande Zhang UTK EECS, 2022–2024
- Oktay Ozturk UTK EECS, 2023–2024

PhD Students

- Kazi Fuad Akhter VCU CS, 2026–Current
- Subarna Saha VCU CS, 2026–Current
- Tianhao Wu UGA SoC, 2023–2026
- Jiawei Yu UGA SoC, 2024–2026
- Yi Chen Liu UGA SoC, 2024–2026

Selected Media Mentions

Jul 2025 **how to poison AI music scrapers** — killswitch@kaleidoscope (Apple Podcast)

- May 2025 **Pitch perfect protection.** *EurekaAlert.*
- Apr 2025 **The Art Of Poison-Pilling Music Files.** Benn Jordan (*YouTube*, **615k** Views).
- Dec 2024 **You can't hear it, but the University of Tennessee tool 'cloaks' songs to protect music from AI.** *Knox News.*
- Oct 2024 **Liu's New Tool Makes Songs Unlearnable to Generative AI.** *UTK News.*
- Oct 2024 **New Tool Makes Songs Unlearnable to Generative AI.** Featured in *TechXplore*, *Softonic*, *Futura*, *Knowridge*, *New Atlas*, etc.
- Oct 2024 **HarmonyCloak slips silent poison into music to corrupt AI copies.** *New Atlas.*
- Oct 2024 **Someone has come up with a cloaking device to fight bogus AI music. It's pretty cool.** *Alan Cross' Journal of Musical Things.*
- Oct 2024 **HarmonyCloak: A New Tool to Protect Musicians from AI Copyright Infringement.** *The Outpost.*
- Oct 2024 **Herramienta dificulta a la IA entrenarse con canciones.** *Tecnología.*
- Oct 2024 **New tech makes songs invisible to AI, protecting artists from copycats.** *Knowledge.*

Professional Activities

- Reviewer, IEEE Computer Society Annual Symposium on VLSI (ISVLSI), (2024).
- Reviewer, IEEE Transactions on Network Science and Engineering, (2024).
- Reviewer, ACM SIGKDD Conference on Knowledge Discovery and Data Mining, (2024)
- Mentor, HackUTK student hackathon, University of Tennessee (2022–2023)
- Volunteer, IEEE Computer Society – Bangladesh Chapter (2017–2019)
- Member and Mentor, MEC Robotics Club, Khulna University of Engineering and Technology, Khulna (2012–2015)